H2 Math Summary

Example Questions

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1 Systems of Linear Equations

1 Three different blends of coffee beans – Blend A, Blend B and Blend C – are sold at both M Café and I Café. The usual selling prices of the blends are the same at both cafés. The total cost of buying one package each of Blend A, Blend B and Blend C is \$176. During a holiday sale the two cafés offered the following discounts:

Café	**Discounts given for each package**			**Total price after the discount**
	Blend A	**Blend B**	**Blend C**	
M Café	15%	10%	5%	\$161.80
I Café	7%	5%	2%	\$169.56

(i) Find the usual selling price of each package of Blend A, Blend B and Blend C coffee beans respectively. [3]

[MI 2025 Prelim P1 Q4]

Solution		**Comments**
(i) **[3]**	Let the usual prices be \$$a$, \$$b$, \$$c$. $a+b+c=176 \quad ---(1)$ $0.85a+0.90b+0.95c=161.80 \quad ---(2)$ $0.93a+0.95b+0.98c=169.56 \quad ---(3)$ Using GC, $a=32,\ b=44,\ c=100$ $\therefore$ Blend A: \$32, Blend B: \$44, Blend C: \$100	

2 A function f is defined by $\mathrm{f}(x)=ax^3+bx^2+cx+d$ where a, b, c and d are constants. The graph of $y=\mathrm{f}(x)$ has a turning point at $x=-1$ and passes through the points $(2, 10)$ and $(-2, -2)$. Given also that $\int_{-2}^{2} \mathrm{f}(x)\,\mathrm{d}x=16$, find the values of a, b, c and d. [4]

[SAJC 2025 Prelim P1 Q1]

Solution		**Comments**
[4]	$\mathrm{f}'(x)=3ax^2+2bx+c$ $\mathrm{f}'(-1)=0$ $3a-2b+c=0 \quad ---(1)$ $\mathrm{f}(2)=10$ $8a+4b+2c+d=10 \quad ---(2)$ $\mathrm{f}(-2)=-2$ $-8a+4b-2c+d=-2 \quad ---(3)$ $\int_{-2}^{2}(ax^3+bx^2+cx+d)\,\mathrm{d}x=16$ $\left[\frac{a}{4}x^4+\frac{b}{3}x^3+\frac{c}{2}x^2+dx\right]_{-2}^{2}=16$ $(4a+\frac{8}{3}b+2c+2d)-(4a-\frac{8}{3}b+2c-2d)=16$ $\frac{16}{3}b+4d=16 \quad ---(4)$ Using GC, $a=3,\ b=0,\ c=-9,\ d=4$	

Two linear equations are independent if and only if neither equation is a linear combination of the other.

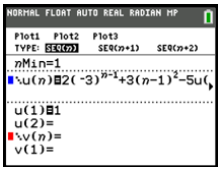
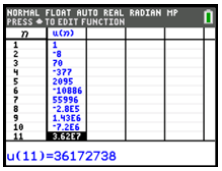
3 A sequence of real numbers, x_n, satisfies the recurrence relation

$$x_{n+1} = a(-3)^n + bn^2 + cx_n, \text{ for } n, a, b, c \in \mathbb{Z} \text{ and } n \geqslant 1.$$

Given that $x_1 = 1$, $x_2 = -8$, $x_3 = 70$, and $x_4 = -377$, find the values of a, b and c. [3]

Hence, find the values of x_{10} and x_{11}. [2]

[HCI 2025 Prelim P1 Q2]

Solution		**Comments**
[2]	$n = 1$ $x_2 = a(-3)^1 + b(1)^2 + cx_1$ $-8 = -3a + b + c \quad ---(1)$ $n = 2$ $x_3 = a(-3)^2 + b(2)^2 + cx_2$ $70 = 9a + 4b - 8c \quad ---(2)$ $n = 3$ $x_4 = a(-3)^3 + b(3)^2 + cx_3$ $-377 = -27a + 9b + 70c \quad ---(3)$ Using GC, $a = 2,\ b = 3,\ c = -5$ $x_{n+1} = 2(-3)^n + 3n^2 - 5x_n$ Using GC, $x_{10} = -7\,210\,868$ $x_{11} = 36\,172\,738$   	

4 The lines l_1 and l_2 have equations

$$l_1: \mathbf{r} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix},$$

$$l_2: \mathbf{r} = \begin{pmatrix} 0 \\ 5 \\ -8 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix},$$

where λ and μ are parameters.

(i) Show that l_1 and l_2 are skew lines. [2]

[SAJC 2025 Prelim P1 Q3]

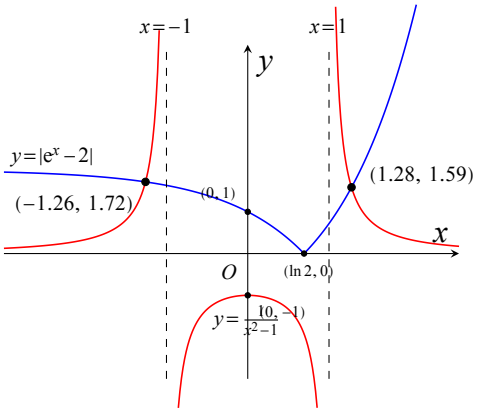
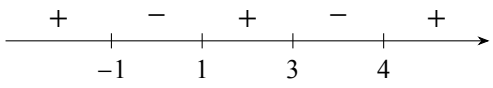
Solution		**Comments**
(i) **[2]**	$1-2\lambda = 2\mu$ $---(1)$ $2+\lambda = 5-2\mu$ $---(2)$ $2\lambda = -8+3\mu$ $---(3)$ Solve (1) and (2): $\lambda = -2,\ \mu = \frac{5}{2}$ Sub into (3): $\text{LHS} = 2(-2) = -4$ $\text{RHS} = -8+3\left(\frac{5}{2}\right) = -\frac{1}{2}$ $\text{LHS} \neq \text{RHS},$ no solution $\therefore l_1$ and l_2 do not intersect $\because \begin{pmatrix} -2 \\ 1 \\ 2 \end{pmatrix} \neq k \begin{pmatrix} 2 \\ -2 \\ 3 \end{pmatrix}$ for any $k \in \mathbb{R}$ $\therefore l_1$ and l_2 are not parallel $\therefore l_1$ and l_2 are skew (shown)	

2 Inequalities

5 **(a)** Solve the inequality $|e^x - 2| > \dfrac{1}{x^2 - 1}$ using a graphical approach. [3]

(b) Without the use of a graphing calculator, solve the inequality $\dfrac{x+2}{x-1} \geqslant \dfrac{2}{x-3}$. [3]

[RVHS 2026 Prelim P1 Q2]

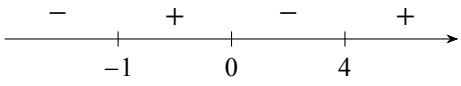
Solution		Comments
(a) **[3]**	 Using GC, intersections at $x=-1.2581$, $x=1.2770$ $\therefore x<-1.26$ or $-1<x<1$ or $x>1.28$	
(b) **[3]**	$\dfrac{x+2}{x-1} - \dfrac{2}{x-3} \geqslant 0$ $\dfrac{(x+2)(x-3)-2(x-1)}{(x-1)(x-3)} \geqslant 0$ $\dfrac{x^2-3x-4}{(x-1)(x-3)} \geqslant 0$ $\dfrac{(x-4)(x+1)}{(x-1)(x-3)} \geqslant 0$  $\therefore x \leqslant -1$ or $1<x<3$ or $x \geqslant 4$	

6 Use an algebraic method to solve the inequalities

$$\frac{x(5x+9)}{x^2+2x+5} > x \geqslant 3x-6.$$

[4]

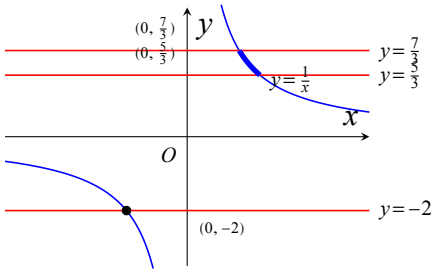
[ACJC 2026 Prelim P1 Q1]

Solution		**Comments**
[4]	$x \geqslant 3x-6$ $x \leqslant 3 \quad ---(1)$ $x^2+2x+5=(x+1)^2+4>0$ for all real x $\frac{x(5x+9)}{x^2+2x+5} > x$ $5x^2+9x > x(x^2+2x+5) \quad (\because x^2+2x+5>0)$ $x^3-3x^2-4x<0$ $x(x-4)(x+1)<0$  $x<-1$ or $0<x<4 \quad ---(2)$ $(1) \cap (2)$: $\therefore x<-1$ or $0<x \leqslant 3$	

7 **(i)** Solve the inequality $3\,|x^2-4| \leqslant |x+2|$. [4]

(ii) Hence, solve $\dfrac{3}{x^2}\,|1-4x^2| \leqslant \left|\dfrac{1+2x}{x}\right|$. [2]

[SAJC 2025 Prelim P2 Q1]

Solution		Comments
(i) **[4]**	$3\|x-2\|\|x+2\| \leqslant \|x+2\|$ $x=-2$: $0 \leqslant 0$ (true) $x \neq -2$: $\|x+2\|>0$ $3\|x-2\| \leqslant 1$ $-\frac{1}{3} \leqslant x-2 \leqslant \frac{1}{3}$ $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$ $\therefore x=-2$ or $\frac{5}{3} \leqslant x \leqslant \frac{7}{3}$	
(ii) **[2]**	$\dfrac{3}{x^2}\,\|1-4x^2\| = 3\left\|\dfrac{1}{x^2}-4\right\| \quad (\because x^2>0)$ $\left\|\dfrac{1+2x}{x}\right\| = \left\|\dfrac{1}{x}+2\right\|$ $3\left\|\left(\dfrac{1}{x}\right)^2-4\right\| \leqslant \left\|\dfrac{1}{x}+2\right\| \quad (x \to \frac{1}{x})$ $\dfrac{1}{x}=-2$ or $\dfrac{5}{3} \leqslant \dfrac{1}{x} \leqslant \dfrac{7}{3}$  $\therefore x=-\frac{1}{2}$ or $\frac{3}{7} \leqslant x \leqslant \frac{3}{5}$	